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$$\int \mathrm{Bgtan}(x)dx$$

$$\begin{cases} u = \mathrm{Bgtan}(x) \\ dv = dx \end{cases} \Rightarrow \begin{cases} du = \dfrac{1}{1+x^2}dx \\ v = x \end{cases}$$

$$= x\,\mathrm{Bgtan}(x) - \int \frac{x}{1+x^2}dx$$

Stel $t = 1 + x^2 \Rightarrow dt = 2xdx$

$$= x\,\mathrm{Bgtan}(x) - \frac{1}{2}\int \frac{dt}{t}$$

$$= x\,\mathrm{Bgtan}(x) - \frac{1}{2}\ln|1+x^2| + c$$

References